Trust and Age Bayesian Digital Twins: Provable Predictive Maintenance under Industrial Network Impairments

Yassir Ameen Al-Karawi, Hamed Al-Raweshidy

***Abstract*—Digital twin predictive maintenance often assumes synchronous, homogeneous measurements, although industrial packets may be delayed, reordered, or dropped. This paper develops TABDT, a trust and age Bayesian twin for remaining useful life (RUL) prediction under geometric delays with a finite deadline and heterogeneous sensor credibility. Arrived packets are aligned in time and weighted by credibility and age before fusion with a random-effects Wiener model. The known-drift scalar anchor limit yields a closed-form steady-state variance bound, two RUL-error scaling regimes, and sufficient conditions for a unique certificate scaled maintenance limit that moves earlier as synchronization worsens. Experiments over 400 simulated units validate the bound within 1.3% and reduce RUL root-mean-square error (RMSE) by up to 27.9%. An external age-compensation test on all 15 XJTU-SY run-to-failure bearings lowers normalized RUL RMSE from 52.11 and 53.02 to 47.70 percentage points. Simulation interval coverage is 93.4% to 95.6%, but external coverage is 76.9%, exposing model mismatch rather than claiming field calibration.**

***Index Terms*—Digital twin, predictive maintenance, industrial cyber-physical systems, remaining useful life, Kalman filtering, age of information, uncertainty quantification, Wiener process.**

I. INTRODUCTION

INDUSTRIAL cyber-physical systems (ICPS) often combine sensing, networking, computing, and control for physical assets. Their value increasingly depends on predicting RUL before an expensive unplanned failure [1], [2]. Digital twins (DTs) support this task by fusing streaming condition-monitoring data with a degradation model [3]. Recent DT-based predictive-maintenance (PdM) studies use synchronized interfaces and homogeneous observation models [4], [5], while trustworthy artificial intelligence (AI) remains central to deployment [2].

Plant networks violate that premise: packets are delayed, reordered, or dropped, while retrofitted sensors differ in calibration and provenance. A twin that treats an old measurement as current can therefore become confidently wrong. Network-oriented DT architectures already identify the communication loop as part of the virtual replica [6]. They do not, however, quantify how ordinary delay and loss propagate into a first-passage RUL decision. Risk frameworks emphasize input quality, robustness, uncertainty measurement, and monitoring [7]. Statistical calibration must be tested against observed errors [8]. Average accuracy alone is insufficient when network degradation invalidates stated confidence.

Intermittent Kalman filtering characterizes estimation under packet loss [9], and age of information (AoI) methods quantify staleness [10]. Communication and analytics codesign has reached condition-based maintenance (CBM) [11], and secure fusion addresses unreliable measurements [12]. Neither supplies first-passage uncertainty under packet age, loss, and calibrated sensor precision. Consequently, existing work does not jointly provide *(G1)* an anchor network-to-RUL certificate, *(G2)* principled use of delayed data, and *(G3)* heterogeneous sensor credibility.

This paper closes that junction with four contributions. *C1)* It formulates Bayesian RUL prediction for a random-effects Wiener process sensed over geometric delays with finite deadlines. *C2)* TABDT compensates an arrived packet for the degradation missed in transit and assigns the variance-matched weight $w_s(a) = \tau_s/(1 + \tau_s aQ/R)$. *C3)* In the known-drift scalar anchor limit, Theorem 1 gives $\bar{P}(p)$. Corollary 1 identifies $p^{-1/4}$ and extreme-loss $p^{-1/2}$ RUL-error scaling, and Proposition 1 maps the certificate to a unique maintenance limit that is monotone with network quality. This scope is distinct from the approximate delayed-data recursion in TABDT. *C4)* Common-random-number experiments over 400 units validate the bound within 1.3% and show up to 27.9% lower RUL RMSE. A single-channel XJTU-SY test retains measured degradation, perturbs only packet delivery, and confirms significant point-error gains while revealing interval undercoverage. Code and raw results are openly available at https://github.com/YassirALKarawi/tabdt-reproducibility and permanently archived as Zenodo release v19, doi: 10.5281/zenodo.21861066.

II. RELATED WORK

A. Digital twins for predictive maintenance: TICPS studies use DT data for bearing diagnosis [13], partial transfer learning [4], and state-space predictive health assessment [5]. Its PHM survey centres RUL and maintenance decisions [1]. None propagates packet age and sensor credibility into a first-passage RUL posterior. A random-effects Wiener model retains inverse Gaussian RUL, unit heterogeneity, and analytic tractability [14], [15], [16].

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B. Imperfect networks and information ageing: Filtering over lossy links provides covariance theory [9]. AoI and out-of-sequence measurement (OOSM) methods describe staleness and delayed updates [10], [17], [18]. Asynchronous Bayesian filtering adapts noise for each source [19], whereas correlated fading makes mean reception probability insufficient under bursty links [20]. These strands do not connect network quality jointly to first-passage RUL and the asymmetric-cost rule used in prognosis-centred CBM [21]. Table I locates TABDT at this missing junction, not as a new degradation law or filter.

III. System Model and Problem Statement

A. Degradation Process

A fleet of nominally identical assets is considered, with a gas-compressor bearing as the running example. Its health index (HI) $x(t) \in \mathbb{R}$ degrades according to a Wiener process with random drift,

$$\begin{aligned} x(t) &= x_0 + \mu t + \sigma_B B(t), \\ \mu &\sim \mathcal{TN}_{[\mu_{\min},\infty)}(\bar{\mu}, \sigma_{\mu 0}^2), \qquad \mu_{\min} > 0. \end{aligned} \tag{1}$$

where $B(t)$ is standard Brownian motion, σ_B is the diffusion coefficient, and $\mathcal{TN}$ denotes a lower-truncated normal parameterized here by the mean and variance of its untruncated parent normal. The positive lower support ensures a proper first-passage model. The random effect on μ captures variability across units caused by manufacturing and load heterogeneity, as is standard in the Wiener prognostics literature [14], [15], [16]. The asset fails when the HI first crosses a threshold D. The failure time $T_f = \inf\{t : x(t) \geq D\}$ then follows, conditionally on μ, an inverse Gaussian (IG) law with mean $(D - x_0)/\mu$ and shape $(D - x_0)^2/\sigma_B^2$ [14]. The true RUL at time t is $\mathrm{RUL}_t = T_f - t$. Sampling at interval Δt and writing $x_k = x(k\Delta t)$ gives the discrete state-space form

$$\mathbf{z}_{k+1} = A\,\mathbf{z}_k + \mathbf{w}_k, \quad \mathbf{z}_k = \begin{bmatrix} x_k \\ \mu_k \end{bmatrix}, \quad A = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \tag{2}$$

with $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathrm{diag}(Q, q_\mu))$, $Q = \sigma_B^2 \Delta t$, and q_μ a small drift random walk that lets the filter track slow load changes.

B. Sensing and Network Model

S condition-monitoring sensors observe the HI,

$$y_k^{(s)} = x_k + v_k^{(s)}, \qquad v_k^{(s)} \sim \mathcal{N}(0,\, R/\tau_s), \quad s = 1, \dots, S. \tag{3}$$

Here R is nominal measurement variance and $\tau_s \in (0, 1]$ is a dimensionless precision score. Equation (3) is the proposed variance calibration map, not a NIST result. The value $\tau_s = 1$ gives nominal variance and lower scores inflate it. From held out filter innovations $r_i^{(s)}$, one can estimate $\widehat{V}_s = \max\{\epsilon_R, N_s^{-1} \sum_i [(r_i^{(s)})^2 - H P_i^- H^\top]\}$ and set $\tau_s = \mathrm{clip}(R/\widehat{V}_s, \tau_{\min}, 1)$. Provenance may regularize this score, consistent with data quality monitoring [7] and robust down-weighting of unreliable channels [12]. This synthetic study specifies τ_s rather than learning it online.

The packet carrying $y_k^{(s)}$ traverses the plant network with raw delay $A_k^{(s)}$ (equivalently, its age on arrival [10]),

$$A_k^{(s)} \sim \mathrm{Geom}_0(p), \qquad a_k^{(s)} = A_k^{(s)} \text{ if } A_k^{(s)} \leq a_{\max}, \tag{4}$$

and is dropped otherwise. Thus $p = \mathbb{P}\{A = 0\}$ is the *timely synchronization probability*, while the deadline-loss probability is $(1 - p)^{a_{\max}+1}$. Delays are independent across packets in the present model and can reorder them, so the twin receives a genuine OOSM stream [17], [18]. The limit $p \to 1$ recovers synchronous sensing. Small p represents congestion or repeated medium-access deferral.

Writing $q = 1 - p$ and $M = a_{\max}$, the delivered mass and its conditional mean age follow directly from the truncated geometric series:

$$\begin{aligned} P_{\mathrm{del}}(p) &= \sum_{a=0}^{M} p q^a = 1 - q^{M+1}, \\ \bar{a}_{\mathrm{del}}(p) &= \mathbb{E}[A \mid A \leq M] = \frac{q[1 - (M+1)q^M + M q^{M+1}]}{p(1 - q^{M+1})}. \end{aligned} \tag{5}$$

Equation (5) separates two network penalties that have different statistical effects. The term $1 - P_{\mathrm{del}}$ removes information, whereas $\bar{a}_{\mathrm{del}}$ makes the information that does arrive stale.

C. Problem Statement

At every step k the twin must deliver (i) a point RUL estimate $\widehat{\mathrm{RUL}}_k$, (ii) a nominal $(1-\alpha)$ approximate prediction interval, and (iii) a preventive maintenance (PM) trigger, using only packets that have physically arrived. Two questions follow. What accuracy can be *guaranteed* as a function of p, and how should stale or low-credibility data be used so that they help rather than mislead? Table II collects the notation.

IV. The TABDT Framework

Fig. 1 shows the architecture. Three design decisions define TABDT, and each answers one of the gaps G1 to G3.

A. Age Compensation with Trust and Age Weighting

A packet $(y_k^{(s)}, k, s)$ arriving at step t with age $a = t - k$ measures the *past* state x_k, which relates to the current state through $x_t = x_k + \mu a \Delta t + \nu_{k:t}$, where $\nu_{k:t} = \sum_{j=k+1}^{t} \varepsilon_j \sim \mathcal{N}(0, aQ)$. After shifting the packet by the estimated drift, its residual with respect to the current state is

$$\tilde{y} - H\mathbf{z}_t = v_k^{(s)} - \nu_{k:t} + a\Delta t(\hat{\mu}_t - \mu), \qquad H = [1\ 0]. \tag{6}$$

Exact OOSM retrodiction retains covariance between state and measurement and between packets [17], [18]. TABDT instead freezes the current drift during a forward shift and neglects shared process noise correlations. Under this first-order conditional-independence approximation, (6) has marginal variance

$$R_{s,a}^{\mathrm{eff}} = \frac{R}{\tau_s} + aQ + (a\Delta t)^2 [P_t]_{\mu\mu}. \tag{7}$$

TABLE I
POSITIONING RELATIVE TO PRIOR WORK

Study / model	Primary prognostic role	Network treatment	Uncertainty and sensor credibility	Decision or guarantee
State space DT health assessment [5]	Synchronized virtual replica and predictive health assessment	Data acquisition in real time is assumed	Model and plant similarity without age dependent sensor credibility	Experimental condition monitoring without an anchor network-to-RUL error law
DT transfer diagnosis [4], [13]	Training data generated by a DT and fault diagnosis across domains	Packet delay, reordering, and deadline loss are not modelled	Domain discrepancy is learned without a provenance score τ_s or stale-packet covariance	Empirical diagnosis without a control limit informed by the network
Adaptive Wiener RUL [16]	Stochastic degradation and online RUL prediction with adaptive drift	Measurements are available to the estimator without an explicit network	Random degradation and parameter uncertainty with a homogeneous observation channel	Probabilistic RUL without a synchronization certificate or PM rule
Intermittent Kalman filtering [9]	State estimation over lossy links	Bernoulli packet arrival or loss	Error-covariance analysis for intermittent observations	Stability and covariance theory without first-passage RUL or maintenance economics
AoI and OOSM methods [10], [17]	Quantify information age or update delayed tracking measurements	Explicit information age and arrival outside sequence	State update that accounts for delay while sensor provenance is outside scope	Timeliness or tracking correction without a DT and PdM decision chain
TABDT (this work)	**Bayesian RUL prediction and preventive maintenance**	**Geometric age, reordering, and finite delivery deadline**	**Joint trust and age weight $w_s(a)$ with decomposed RUL variance $\sigma^2_{\mathrm{RUL},t}$**	**Closed form $\bar{P}(p)$, two scaling regimes, and $\xi^*(p)$ that is monotone with network quality**

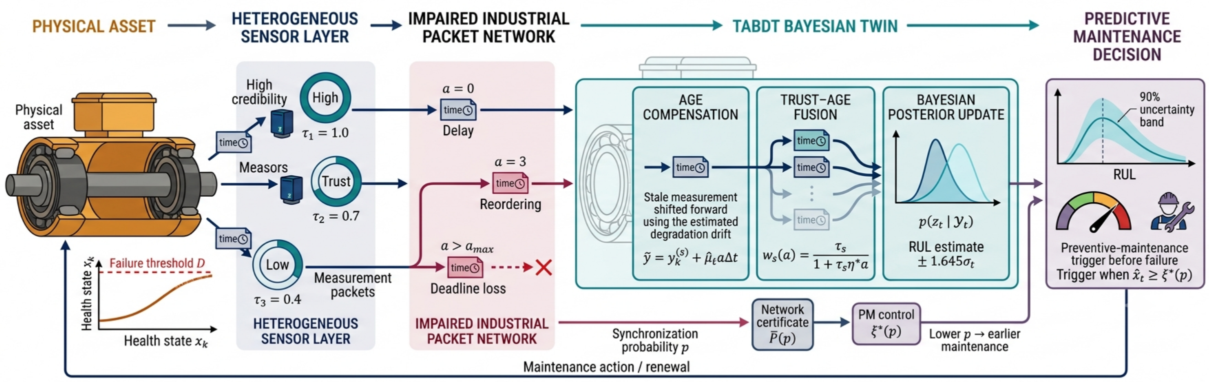

Fig. 1. End-to-end TABDT architecture. Timestamped sensor observations undergo delay, reordering, and deadline loss. Age compensation, trust–age fusion, and Bayesian updating yield state and RUL posteriors. The certificate $\bar{P}(p)$ sets the preventive-maintenance limit $\xi^*(p)$.

TABLE II
PRINCIPAL NOTATION

Symbol	Meaning
x_k, D	health index at step k, failure threshold
μ, $\bar{\mu}$, $\sigma_{\mu 0}$	unit drift, fleet mean, heterogeneity
σ_B, Q	diffusion coefficient and process variance $\sigma_B^2 \Delta t$
$y_k^{(s)}$, R, τ_s	measurement, nominal variance, sensor precision
$a_k^{(s)}$, p, q, $a_{\max}$	packet age, on time probability, $1-p$, deadline
$w_s(a)$, η^*	trust and age weight with $\eta^* = Q/R$
P_{del}	probability of delivery before the deadline
$\hat{\mathbf{z}}_k$, P_k	state estimate and error covariance
$\bar{P}(p)$	steady-state variance bound (Thm. 1)
ξ, ξ^*	PM control limit and its optimum (Prop. 1)
c_p, c_f	preventive / failure replacement cost

This motivates shifting the measurement by missed degradation and charging the retained marginal uncertainties to its likelihood:

$$\tilde{y} = y_k^{(s)} + \hat{\mu}_t\, a\Delta t, \tag{8}$$

$$\tilde{R} = \frac{R}{w} + (a\Delta t)^2\, [P_t]_{\mu\mu}, \qquad w = w_s(a) = \frac{\tau_s}{1+\tau_s \eta a}. \tag{9}$$

Sequential use of (8) and (9) is therefore approximate, not the exact delayed data posterior. Sensitivity and coverage tests expose the practical effect of omitted correlations. Trust τ_s sets variance for a fresh packet, while age a reduces delayed information. Because $R/w_s(a) = R/\tau_s + \eta a R$, the choice

$$\eta^* = Q/R \tag{10}$$

makes $R/w_s(a) = R/\tau_s + aQ$, the sensor variance plus Wiener variance accrued in transit. Thus the age rate is the ratio of process variance to measurement variance, with monotonicity

$$\frac{\partial w_s}{\partial a} = -\frac{\tau_s^2 \eta}{(1+\tau_s\eta a)^2} < 0, \qquad \frac{\partial w_s}{\partial \tau_s} = \frac{1}{(1+\tau_s\eta a)^2} > 0. \tag{11}$$

Section VI examines this ratio over four orders of magnitude. The additive term in (9) charges uncertainty in drift estimation. Omitting it overweights aged packets while drift is poorly known.

Each packet delivered within the deadline is used exactly once on arrival. Replaying a stored value would count the same information twice.

B. Bayesian Recursion

With (8) and (9), the twin runs a sequential Kalman recursion on the state (2). A time update occurs at every step, followed by one scalar measurement update per arriving packet with observation matrix $H = [1\ 0]$, constructed measurement $\tilde{y}$, and noise $\tilde{R}$. Algorithm 1 lists the complete procedure. The

Algorithm 1 Trust and age Bayesian recursion at time t

Require: posterior $(\hat{\mathbf{z}}_{t-1}, P_{t-1})$, unused arrived packets $\mathcal{A}_t$, and $Q, q_\mu, R, \{\tau_s\}, \eta^*$
Invariant: each packet is fused once, in nondecreasing age $a = t - k$ (newest first).

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Predict: ẑ ← A ẑ_{t-1}, P ← A P_{t-1} A^⊤ + diag(Q, q_μ).
Sort A_t by (a, s). For each (y_k^{(s)}, k, s) ∈ A_t do
    a ← t − k and w ← τ_s/(1 + τ_s η* a).
    ỹ ← y_k^{(s)} + μ̂ aΔt and R̃ ← R/w + (aΔt)^2 P_{μμ}.
    K ← P H^⊤ (H P H^⊤ + R̃)^{-1} and e ← ỹ − H ẑ.
    ẑ ← ẑ + K e, J ← I − K H, and P ← J P J^⊤ + K R̃ K^⊤.
End for. Symmetrize P ← (P + P^⊤)/2.
Evaluate (13) through (16). Set u_t ← 1{x̂_t ≥ ξ*(p)}.
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Ensure: $\hat{\mathbf{z}}_t, P_t, \widehat{\mathrm{RUL}}_t \pm z_{1-\alpha/2}\sigma_{\mathrm{RUL},t}$, and PM flag u_t.

prior on the drift, $\hat{\mu}_0 = \bar{\mu}$ with variance $\sigma^2_{\mu 0}$, is a moment-matched Gaussian approximation to the lower-truncated fleet random effect in (1). The recursion therefore performs drift learning for each unit in the sense of [16] as a consequence of ordinary filtering.

C. RUL Posterior

Given $\hat{\mathbf{z}}_t = [\hat{x}_t, \hat{\mu}_t]^\top$ and P_t, let $m_t = \max(D - \hat{x}_t, \epsilon_x)$ and $\hat{\mu}_t^+ = \max(\hat{\mu}_t, \epsilon_\mu)$. Drift is locally frozen beyond t. This treatment is exact for $q_\mu = 0$ and approximate for the slow random walk in (2). Then the first-passage RUL $L_t = T_f - t$ has the inverse Gaussian density

$$f_{L_t|x_t,\mu}(\ell) = \frac{D - x_t}{\sqrt{2\pi\sigma_B^2 \ell^3}} \exp\left[-\frac{(D - x_t - \mu\ell)^2}{2\sigma_B^2 \ell}\right], \quad \ell > 0. \tag{12}$$

Equation (12) has conditional mean $(D - x_t)/\mu$ and variance $(D - x_t)\sigma_B^2/\mu^3$. The point estimate is therefore the conditional mean evaluated at the posterior estimates,

$$\widehat{\mathrm{RUL}}_t = \frac{m_t}{\hat{\mu}_t^+}. \tag{13}$$

For $g(x, \mu) = (D - x)/\mu$, the gradient at the posterior mean is

$$\nabla g_t = \begin{bmatrix} -(\hat{\mu}_t^+)^{-1} & -m_t(\hat{\mu}_t^+)^{-2} \end{bmatrix}^\top. \tag{14}$$

Applying the law of total variance and the first-order delta method gives

$$\mathrm{Var}(L_t \mid \mathcal{Y}_t) \approx \mathbb{E}[\mathrm{Var}(L_t \mid \mathbf{z}_t) \mid \mathcal{Y}_t] + \nabla g_t^\top P_t \nabla g_t. \tag{15}$$

Substituting (14) into (15) separates the aleatoric first-passage part and the epistemic state-estimation part [14]:

$$\sigma^2_{\mathrm{RUL},t} \approx \underbrace{\frac{m_t \sigma_B^2}{(\hat{\mu}_t^+)^3}}_{\text{aleatoric (IG)}} + \underbrace{\frac{[P_t]_{xx}}{(\hat{\mu}_t^+)^2} + \frac{m_t^2 [P_t]_{\mu\mu}}{(\hat{\mu}_t^+)^4} + \frac{2m_t [P_t]_{x\mu}}{(\hat{\mu}_t^+)^3}}_{\text{epistemic (delta method)}}. \tag{16}$$

The nominal Gaussian delta-method interval is $\widehat{\mathrm{RUL}}_t \pm z_{1-\alpha/2}\sigma_{\mathrm{RUL},t}$. It is not an exact Bayesian credible interval because (16) plugs in the expected aleatoric term and omits higher posterior moments. Calibration must therefore be tested [8]. The expansion retains the covariance between state and drift required by the delta method. Theorem 1 bounds the $[P_t]_{xx}$ contribution for the anchor-filter limit stated next. It does not claim to bound the OOSM approximation with unknown drift in Algorithm 1.

V. Theoretical Guarantees

A. A Closed-Form Variance Certificate

For analysis, an anchor with unit trust uses the known-drift scalar case, retaining fresh packets and omitting delayed ones. This reference is independent of Algorithm 1's OOSM approximation. Let P_k denote the prior variance one step ahead for this intermittent Kalman filter, which receives a fresh measurement with probability p per step [9]. Its Riccati recursion is

$$P_{k+1} = P_k + Q - \gamma_k \frac{P_k^2}{P_k + R}, \qquad \gamma_k \sim \mathrm{Bern}(p). \tag{17}$$

Theorem 1 (Steady state variance bound)**.** *For the marginally stable degradation dynamics (2) observed through (17) with any $p \in (0, 1]$,*

$$\limsup_{k\to\infty} \mathbb{E}[P_k] \;\le\; \bar{P}(p) \;=\; \frac{Q + \sqrt{Q^2 + 4pQR}}{2p}. \tag{18}$$

Proof. Write $g(P) = P^2/(P + R)$. Since $g''(P) = 2R^2/(P + R)^3 > 0$, g is convex on $[0, \infty)$, and Jensen's inequality gives $\mathbb{E}[g(P_k)] \ge g(\mathbb{E}[P_k])$. Taking expectations of (17),

$$\mathbb{E}[P_{k+1}] = \mathbb{E}[P_k] + Q - p\,\mathbb{E}[g(P_k)] \;\le\; f\big(\mathbb{E}[P_k]\big),$$

with $f(u) = u + Q - p\,g(u)$. Because $g'(u) = (u^2 + 2uR)/(u + R)^2 \in [0, 1)$, the derivative $f'(u) \in (1 - p, 1]$. Hence f is nondecreasing, and the comparison sequence $u_{k+1} = f(u_k)$, $u_0 = \mathbb{E}[P_0]$, dominates $\mathbb{E}[P_k]$ for all k. The fixed points of f solve $Q = p\,g(u)$, i.e., $pu^2 - Qu - QR = 0$, whose unique positive root is $\bar{P}(p)$ in (18). Since $f(u) - u = Q - p\,g(u)$ is positive below $\bar{P}(p)$ and negative above it, and f is monotone, $u_k \to \bar{P}(p)$ from any initialization, which yields the claim. □

Implicit differentiation of $p\bar{P}^2 = Q(\bar{P} + R)$ gives

$$\frac{d\bar{P}}{dp} = -\frac{\bar{P}^2}{2p\bar{P} - Q} = -\frac{\bar{P}^2}{\sqrt{Q^2 + 4pQR}} < 0. \tag{19}$$

Thus (19) proves that the certificate worsens strictly as synchronization probability falls.

Corollary 1 (Two network scaling regimes)**.** *For the known-drift anchor filter, let $\delta = Q/(pR)$. If $\delta \to 0$ (equivalently, $p \gg Q/R$), then*

$$\begin{aligned} \bar{P}(p) &= \sqrt{\frac{QR}{p}}\,[1 + O(\sqrt{\delta})], \\ \sigma^{\mathrm{epi}}_{\mathrm{RUL}} &\lesssim \frac{(QR)^{1/4}}{\mu} p^{-1/4}[1 + O(\sqrt{\delta})]. \end{aligned} \tag{20}$$

In the extreme-loss limit $pR/Q \to 0$,

$$\bar{P}(p) = \frac{Q}{p} + R + O\left(\frac{pR^2}{Q}\right), \qquad \sigma^{\mathrm{epi}}_{\mathrm{RUL}} \lesssim \frac{\sqrt{Q}}{\mu} p^{-1/2}. \tag{21}$$

Here $Q/R = 9 \times 10^{-4}$, whereas the evaluated range has $p \ge 0.02$, so $\delta \le 0.045$ and (20) is the relevant approximation. Within that regime, halving p increases the leading epistemic RUL scale by $2^{1/4} - 1 \approx 18.9\%$. The true $p \to 0$ exponent is nevertheless $-1/2$ by (21). TABDT can perform better by exploiting delayed packets. The anchor result supplies only a network scale reference for the surrogate below.

B. Maintenance Threshold Informed by the Network

The certificate becomes operational when it prices maintenance risk. Consider the classical control-limit policy [21]: replace preventively (cost c_p) when the *estimate* $\hat{x}$ crosses $\xi < D$. An undetected crossing of D before the trigger costs $c_f > c_p$. Let $e_k = x_k - \hat{x}_k$ denote the zero-mean anchor-filter error. An unnoticed failure requires e_k to exceed the safety margin $m = D - \xi$. Theorem 1 gives the distribution-free Cantelli bound

$$\limsup_{k\to\infty} \mathbb{P}\{e_k \geq m\} \leq \frac{\bar{P}(p)}{\bar{P}(p) + m^2}, \qquad m > 0. \tag{22}$$

Because variance alone does not imply a Gaussian tail, a smooth Gaussian surrogate that can be tested by simulation is defined as

$$C_G(\xi, p) \triangleq \left[c_p + (c_f - c_p)\,\bar{\Phi}\Big(\tfrac{D-\xi}{\sqrt{\bar{P}(p)}}\Big)\right] \frac{\mu}{\xi}, \tag{23}$$

where $\bar{\Phi}$ and ϕ are the standard normal tail and density. The factors approximate risk from a late trigger and renewal frequency, respectively. (23) is not a probabilistic upper bound, unlike (22).

Proposition 1 (Unique control limit and network monotonicity)**.** *Let $s = \sqrt{\bar{P}(p)}$ and $d = c_f - c_p > 0$. If*

$$\frac{Dd}{s\sqrt{2\pi}} > \frac{c_f + c_p}{2}, \tag{24}$$

the surrogate (23) has a unique interior minimizer $\xi^* \in (0, D)$ characterized by

$$(c_f - c_p)\,\frac{\xi^*}{\sqrt{\bar{P}(p)}}\,\phi\Big(\tfrac{D-\xi^*}{\sqrt{\bar{P}(p)}}\Big) = c_p + (c_f - c_p)\,\bar{\Phi}\Big(\tfrac{D-\xi^*}{\sqrt{\bar{P}(p)}}\Big). \tag{25}$$

Let $r^* = (D-\xi^*)/s$. Wherever $(r^*)^2\xi^* > D$, the safety margin $D - \xi^*$ is strictly increasing in s and hence nonincreasing in p.

Proof. Write $h(\xi) = c_p + d\bar{\Phi}((D-\xi)/s)$, so $C'_G(\xi, p)$ has the sign of $F(\xi) = \xi h'(\xi) - h(\xi)$. Here $F(0) < 0$, condition (24) is exactly $F(D) > 0$, and $F'(\xi) = \xi h''(\xi) > 0$ for $0 < \xi < D$. Hence the root is unique and gives (25). Implicit differentiation of (25) yields $d(D-\xi^*)/ds = r^* - D/(r^*\xi^*)$, which is positive under the stated condition. Finally, s decreases with p by (18). □

Equation (23) uses a Gaussian tail and a first-order renewal scale and ignores diffusion overshoot near D. Section VI tests these approximations. Define the stationary point residual $F_p(\xi) = \xi h'(\xi) - h(\xi)$ from the proof of Proposition 1. Because F_p is strictly increasing when (24) holds, Algorithm 2 uses a bracketed root solve rather than an unconstrained optimizer. The returned solution therefore inherits the uniqueness guarantee.

C. Computational Complexity

Let $m_t = |\mathcal{A}_t|$. One step of Algorithm 1 costs $O(n^3 + m_t n^2 + m_t \log m_t)$ operations and $O(n^2 + m_t)$ memory. In steady state $\mathbb{E}[m_t] = SP_{\text{del}} \leq S$, whereas the deadline gives $m_t \leq S(a_{\max}+1)$. Hence $O(Sn^2)$ for fixed $n = 2$ is expected, not a worst-batch bound. Algorithm 2 is one $O(\log(D/\epsilon))$ solve. This count is not a claim about operation in real time. Platform latency and energy still require profiling.

Algorithm 2 PM threshold informed by the network

Require: $p, Q, R, D, \mu, c_p, c_f$ and root tolerance $\epsilon > 0$.

1: $\bar{P} \leftarrow (Q + \sqrt{Q^2 + 4pQR})/(2p)$, $s \leftarrow \sqrt{\bar{P}}$, and $d \leftarrow c_f - c_p$.
2: $h(\xi) \leftarrow c_p + d\bar{\Phi}((D-\xi)/s)$ and $F_p(\xi) \leftarrow \xi d\phi((D-\xi)/s)/s - h(\xi)$.
3: **if** $F_p(D) \leq 0$ **then** compare the admissible boundary costs and return the boundary with lower cost.
4: **else** solve $F_p(\xi) = 0$ on $(0, D)$ by a safeguarded bracketed method (bisection/Brent) until the bracket width is at most ϵ.
5: Set ξ^* to the bracketed root. Evaluate $C_G(\xi^*, p)$ from (23).

Ensure: unique $\xi^*(p)$ when (24) holds and surrogate cost $C_G(\xi^*, p)$.

TABLE III
SIMULATION CONFIGURATION

Parameter	Value	Parameter	Value
Δt	1 h	D	10 HI
$\bar{\mu}$	0.02 HI/h	$\sigma_{\mu 0}$	0.004 HI/h
$\mu_{\min}$	0.01 HI/h	P_0	$\mathrm{diag}(0.25, \sigma_{\mu 0}^2)$
σ_B	0.03 HI/$\sqrt{\text{h}}$	Q	9×10^{-4}
R	1.0	(τ_1, τ_2, τ_3)	(1.0, 0.7, 0.4)
q_μ	10^{-7}	$\eta^* = Q/R$	9×10^{-4}
$a_{\max}$	100 steps	(c_p, c_f)	(1, 10)
Monte Carlo units	400	eval. window	$[0.2,\ 0.9]\,T_f$

VI. Numerical Study

A. Setup and Physics Validation

Table III specifies a slow gas-compressor bearing degradation process ($\bar{\mu} = 0.02$ HI/h, mean life $D/\bar{\mu} = 500$ h) monitored by $S = 3$ sensors of unequal trust through a network. The probability p is varied from 0.02 under harsh conditions to 1.0 under ideal conditions. At $p = 0.02$ and $a_{\max} = 100$, the implied deadline-loss probability is 13.0%. Before any comparison, the simulator itself was validated against the closed-form result. Over 20 000 paths with fixed drift, the empirical mean failure time was 501.3 h against the IG value of 500.0 h, and the empirical standard deviation 33.6 h against 33.5 h. This physics based check is treated as a precondition for trusting everything that follows.

B. Baselines and Metrics

Four estimators share identical truths, noises, and packet realizations, differing only in how packets enter the update: *B1* is the open-loop twin with only the model, fleet-mean drift, and no data. *B2* is a deliberately naive ablation that applies every arriving packet as if its age were zero and is not asserted to represent all DT implementations. *B3* is the intermittent Kalman filter that uses only packets arriving on time and discards the rest, namely the policy characterized by Theorem 1 [9]. *TABDT* is Algorithm 1. The reported measures are RUL RMSE and the 90% coverage probability of the prediction interval (PICP) over the $[20\%, 90\%]$ life window of each unit. The same truths, normal measurement draws, and delay uniforms obtained through the inverse CDF are reused across all p values. Vertical error bars are ± 1.96 standard errors across units. For the evaluation index set

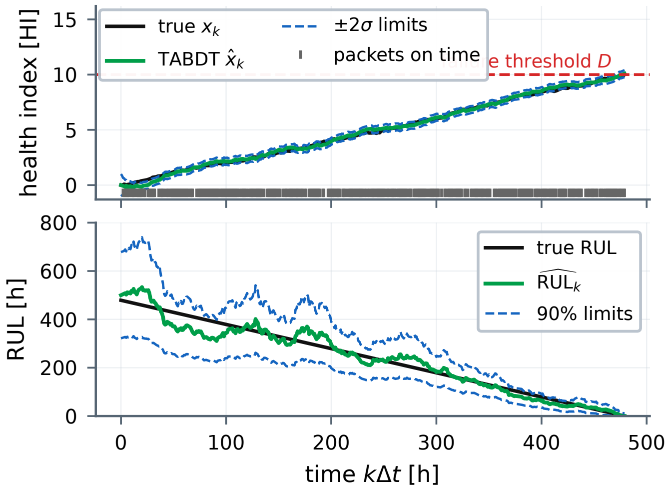

Fig. 2. One realization at $p = 0.4$. Top: true and estimated health index with paired $\pm 2\sigma$ limits, failure threshold D, and packets arriving on time (rug marks). Bottom: true RUL against the TABDT estimate (13) with paired 90% limits from (16).

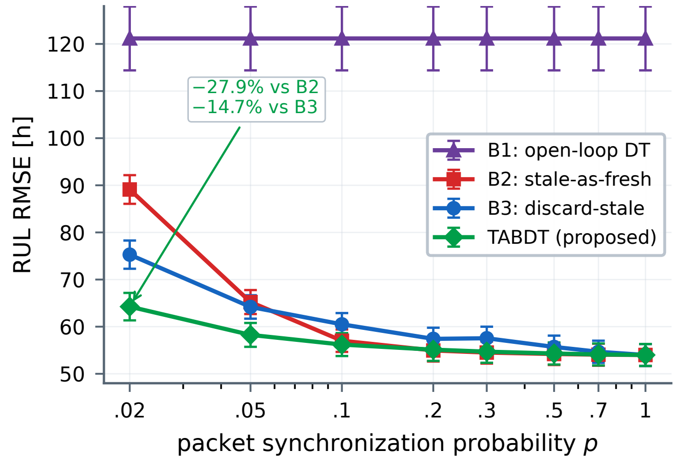

Fig. 3. RUL RMSE from (26) against synchronization probability p for packets arriving on time. The abscissa is logarithmic and vertical bars show 95% confidence intervals over 400 units. All methods driven by data coincide at $p = 1$. The annotation quantifies the gain of TABDT at $p = 0.02$, where staleness and deadline loss are most severe.

$\mathcal{I} = \{(n,k) : 0.2T_f^{(n)} \le k\Delta t \le 0.9T_f^{(n)}\}$, the two reported metrics are

$$\begin{aligned} e_{n,k} &= \widehat{\mathrm{RUL}}_{n,k} - \mathrm{RUL}_{n,k}, \\ \mathrm{RMSE}_{\mathrm{RUL}} &= \left[|\mathcal{I}|^{-1} \sum_{(n,k)\in\mathcal{I}} e_{n,k}^2 \right]^{1/2}, \\ \mathrm{PICP}_{0.9} &= |\mathcal{I}|^{-1} \sum_{(n,k)\in\mathcal{I}} \mathbf{1}\{|e_{n,k}| \le 1.645\,\sigma_{\mathrm{RUL},n,k}\}. \end{aligned} \tag{26}$$

Thus $\mathrm{RMSE}_{\mathrm{RUL}}$ measures point accuracy, whereas $\mathrm{PICP}_{0.9}$ tests whether the uncertainty decomposition in (16) attains its stated empirical coverage.

C. Results

1) What the twin sees: Fig. 2 shows a single unit at $p = 0.4$. Timely arrivals are sparse, the estimate coasts between updates, and the RUL posterior contracts toward the truth as drift learning proceeds, with paired 90% limits from (16).

2) Accuracy versus network quality: Fig. 3 is the central comparison and Table IV extracts three operating points. All methods driven by data coincide at $p = 1$ (54.0 h) because no delayed packet is available to distinguish their update rules. Common random numbers keep B1 exactly constant at 121.2 h, the price of never learning unit-specific drift. As p falls, B3 pays for discarding usable packets. Its RMSE rises to 64.2 h at $p = 0.05$ and 75.3 h at $p = 0.02$. B2 is competitive under moderate delay but becomes biased under severe staleness, reaching 65.2 and 89.1 h at the same operating points. TABDT remains at 58.3 h for $p = 0.05$ and 64.3 h for $p = 0.02$. Thus, at $p = 0.02$, age compensation and trust and age weighting reduce RMSE by 14.7% relative to B3 and 27.9% relative to B2. At $p = 0.05$, the reductions are 9.2% and 10.7%. For $p \ge 0.05$, TABDT remains between 54.0 h and 58.3 h, so a twentyfold drop in timely arrival costs 4.3 h here.

Conditioned on delivery before $a_{\max} = 100$, (4) gives a mean packet age of 33.9 steps at $p = 0.02$. An update that

TABLE IV
HEADLINE COMPARISON AT THREE NETWORK CONDITIONS

Method	$p = 0.02$		$p = 0.05$		$p = 0.20$	
	RMSE [h]	PICP [%]	RMSE [h]	PICP [%]	RMSE [h]	PICP [%]
B1 open-loop	121.2	95.0	121.2	95.0	121.2	95.0
B2 stale-as-fresh	89.1	91.5	65.2	97.0	**55.0**	96.4
B3 discard-stale	75.3	95.7	64.2	95.8	57.4	95.9
TABDT (proposed)	**64.3**	93.4	**58.3**	94.5	55.1	95.2

treats stale data as fresh therefore omits, on average, $\bar{\mu}\,\mathbb{E}[a \mid a \le a_{\max}]\Delta t \approx 0.68$ HI of degradation, explaining the rapid B2 error increase. TABDT adds this missing drift through (8) and simultaneously charges the diffusion term aQ and drift uncertainty $(a\Delta t)^2 P_{\mu\mu}$ through (9). B3 avoids the bias but sacrifices all delayed information.

3) Empirical interval coverage: Fig. 4 evaluates the statistical calibration question emphasized in RUL uncertainty studies [8]. With the full state and drift cross-covariance retained in (16), none of the methods undercovers in this experiment. TABDT attains coverage between 93.4% and 95.6%, while B3 produces coverage between 95.5% and 95.9%. B2 is closest to nominal at the single harshest point (91.5% at $p = 0.02$) but overcovers by 5.6 to 7.2 percentage points elsewhere. These intervals are therefore conservative, not perfectly calibrated, and the point RMSE gain in Fig. 3 should not be read as exact posterior calibration.

4) Tightness of the certificate: Fig. 5 confronts Theorem 1 with Monte Carlo estimates from 4 000 paths for the steady-state prior variance and the actual mean square estimation error. The bound holds at every p and is remarkably tight. The ratio $\bar{P}(p)/\mathbb{E}[P]$ ranges from 1.000 at $p = 1$ to 1.013 at $p = 0.05$, so the Jensen slack in the proof costs at most 1.3% in this regime. The dotted $\sqrt{QR/p}$ line confirms the moderate-loss slope of $-1/2$ for the variance over the evaluated range. It is not the $p \to 0$ asymptote, where (21) predicts slope -1 below the crossover $p \approx Q/R$. Proposition 1 uses the exact bound.

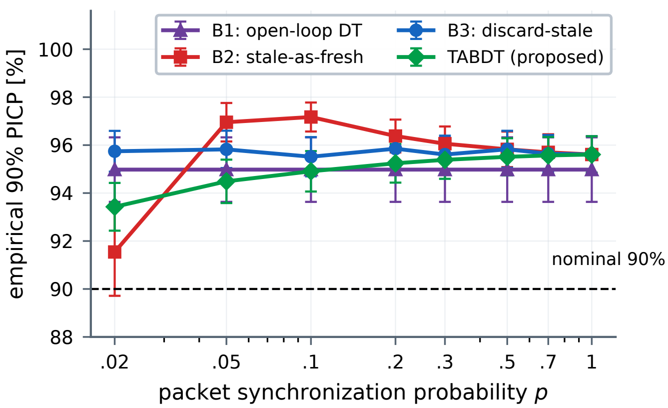

Fig. 4. Empirical coverage $\mathrm{PICP}_{0.9}$ from (26). The abscissa is logarithmic and vertical bars show 95% confidence intervals across units. The dashed line is the nominal 90% target. TABDT remains conservative from 93.4% to 95.6%.

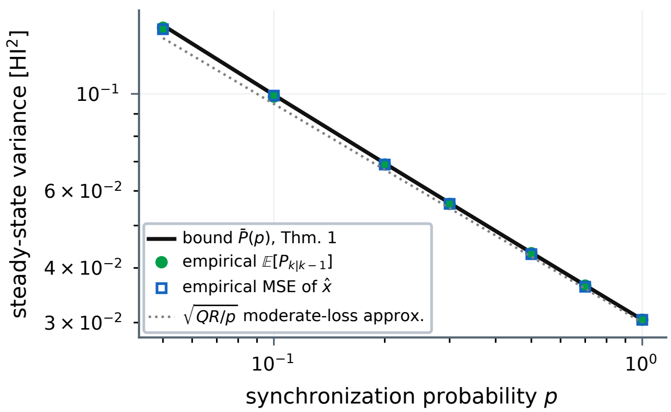

Fig. 5. Validation of Theorem 1. The closed-form bound $\bar{P}(p)$ exceeds both the empirical prior variance and the empirical MSE at every p, with at most 1.3% slack, and follows the $\sqrt{QR/p}$ moderate-loss approximation over this range.

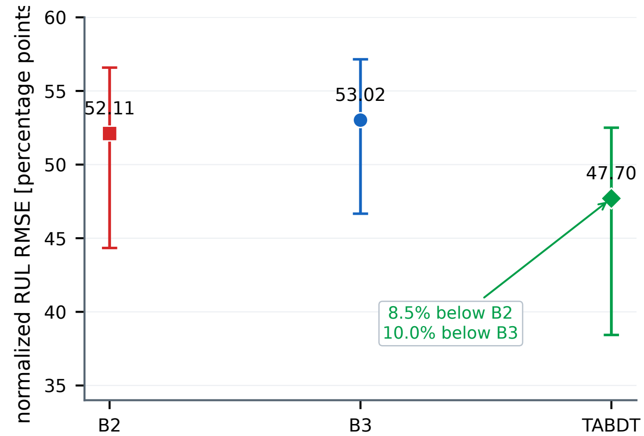

Fig. 6. External XJTU-SY age-compensation test at $p = 0.02$. Bars are 95% cluster-bootstrap intervals over 15 bearings. Network repetitions are not treated as independent specimens.

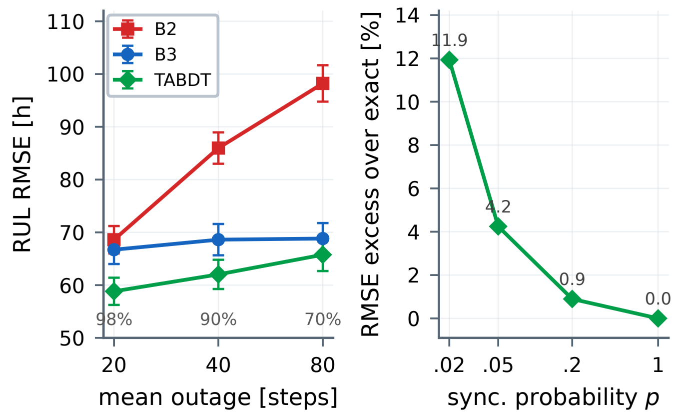

Fig. 7. Stress tests. Left: RUL RMSE under a two-state Markov link at timely fraction 0.05 as the mean outage lengthens, with common channel uniforms (95% confidence intervals, delivered fractions below). Right: TABDT RMSE excess over exact reprocessing of all arrived packets on a common 25-step grid.

5) Sensitivity and external run-to-failure validation: The prespecified η/η^* sweep changes RMSE by at most 6.5%. At $\eta/\eta^* = 1$ versus 100, errors are 54.7 versus 51.2 h at $p = 0.05$, with smaller shifts at $p = 0.2$ and 0.5.

A single-channel XJTU-SY test uses 15 bearing lives under three operating conditions [22]. The shared causal proxy $\mathrm{clip}[(\mathrm{peak}/b - 1)/9, 0, 1.2]$ maps a condition baseline b to zero and $10b$ to one. It is a normalization, not a failure label. Within-condition leave-one-bearing-out calibration uses $m = 4$ lives and residual variance $s_w^2 + (1 + 1/m)s_b^2$. Thus 1.25 is derived for a new bearing relative to the calibration mean, not tuned to coverage. Forty common packet realizations use $p = 0.02$ and a 30-record deadline. All 15 lives form the primary aggregate. The original protocol prespecified the $\geq$ 100-record cohort (13 lives), retained here as sensitivity. Between arrivals, TABDT coasts on calibrated drift, whereas B2 and B3 hold their last accepted value.

Fig. 6 gives normalized RUL RMSE. TABDT reaches 47.70 points, versus 52.11 when delayed records are treated as current and 53.02 when they are discarded, reductions of 8.5% and 10.0%. Bearing-clustered paired 95% bootstrap intervals are 3.59 to 6.72 and 4.12 to 9.56 points, excluding zero. The 13-life sensitivity returns 47.77, 52.16, and 53.01 points, with 76.7% coverage. The nominal 90% interval covers 76.9% over all 15 lives, while factors 1 to 2 give 72.4% to 84.5% in the primary aggregate. Thus point error supports age compensation, but not field calibration.

6) Stress beyond the design assumptions: Fig. 7 reports two checks on the same 400-unit population. A two-state Markov link at timely fraction 0.05 with mean outages of 20 to 80 steps preserves the method ordering under bursts that the iid certificate does not cover [20]. Exact reprocessing gives the exact posterior that the forward correction approximates, and TABDT exceeds it by 11.9% at $p = 0.02$, 0.9% at $p = 0.2$, and zero at $p = 1$, versus $O(Stn^2)$ work per evaluation at time t. TABDT has the online cost stated above.

7) From certificate to maintenance decisions: Fig. 8 closes the loop from network quality to money. The surrogate (23) with $\bar{P}(p)$ from Theorem 1 is plotted against ξ for three network conditions and overlaid with independent Monte Carlo policy simulations. Three observations. First, analytic and simulated cost rates agree within 0.6% for sampled points with $\xi \leq 9.0$. Second, it degrades exactly where the derivation predicts. Near D, diffusion overshoot ignored by (23) raises the simulated rate by 7.3% at $(p, \xi) = (0.2, 9.5)$. The design

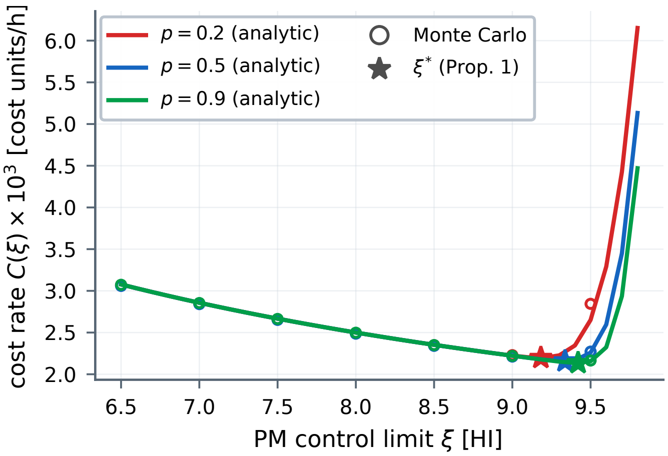

Fig. 8. Long run maintenance cost rate against the PM limit ξ. Lines show (23), circles show independent Monte Carlo results, and stars mark ξ^*. Poorer synchronization moves ξ^* toward earlier maintenance. Approximation error is largest near D.

rule should not be trusted for margins thinner than $\sqrt{\bar{P}(p)}$, which the certificate supplies. Third, the surrogate optimum moves with the network as Proposition 1 requires: $\xi^* = 9.42$ at $p = 0.9$, 9.34 at $p = 0.5$, and 9.18 at $p = 0.2$. The safety margin widens from 0.58 to 0.82 HI and the minimum cost rises from 2.135 to 2.197 $\times 10^{-3}$ per hour. Both sufficient conditions of Proposition 1 hold at these points (smallest $(r^*)^2\xi^*$ 88.5 against $D = 10$).

D. Threats to Validity and Limitations

Theorem 1 covers linear Gaussian Wiener degradation with independent delays, and the Markov-outage test probes that boundary only empirically [20]. The main study is synthetic with fixed τ_s. XJTU-SY tests only the injected packet-use rule on one peak index from 15 accelerated lives. The forward OOSM correction omits correlations retained by exact retrodiction [17], [18], and its measured excess over exact reprocessing is empirical, not proven. Frozen-drift intervals cover only 76.9% externally, (23) remains a design surrogate, and correlated fading, nonlinear degradation, edge profiling, and plant trials remain open.

VII. Conclusion

TABDT aligns network-impaired packets, encodes trust and age in covariance, and propagates uncertainty through $\bar{P}(p)$ to the maintenance threshold. Simulation places the bound within 1.3%, with up to 27.9% lower RUL RMSE and 93.4% to 95.6% coverage. The 15-life XJTU-SY test reaches 47.70 points versus 52.11 and 53.02, with 76.9% coverage exposing calibration limits for plant trials.

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